

EEE283

Digital Logic Design

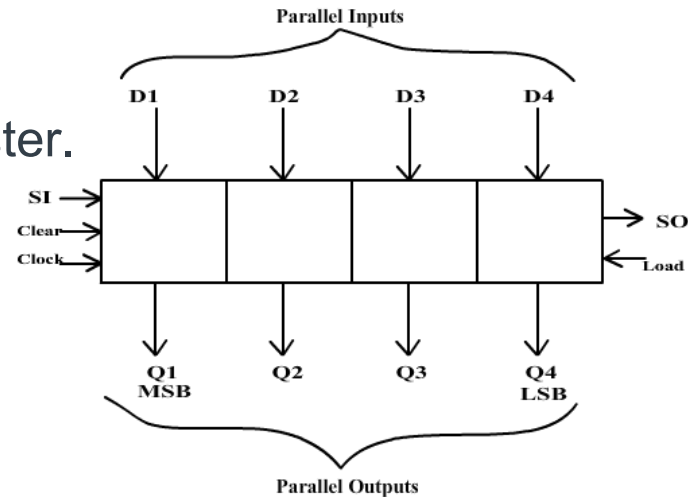
Course Instructor: A K M Anindya Alam

Topic 9:

Sequential Circuits 2 (Registers and Counters)

Registers

- A register is a group of flip-flops that can store multi-bit data.
- A flip-flop stores one bit of information. When a set of n flip-flops is used to store n bits of information, such as an n -bit number, we refer to the group of flip-flops as a register.
- A common clock is used for each flip-flop in a register.



Load control in Registers

- A register operates based on the frequency of the clock signal.
- As a result, the data is stored only for one time period after which it may be overwritten or replaced by undefined (garbage) values.
- **To prevent unintentional data loss, an independent external load control is introduced.**
- This ensures that data is loaded or updated only when required, preserving the stored value until a new valid input is given.

Classification of Registers

I. CLASSIFICATION BASED ON I/P & O/P

1 SISO (Serial Input Serial Output)



- Data shifted in one bit at a time.
- Data shifted out one bit at a time.
- **Typical application:** Delay line.

2 SIPO (Serial Input Parallel Output)



- Data entered bit-by-bit.
- Data available on all output lines simultaneously.
- **Typical application:** Serial-to-parallel converter.

3 PISO (Parallel Input Serial Output)



- Data loaded into all stages simultaneously.
- Data shifted out bit-by-bit.
- **Typical application:** Parallel-to-serial converter.

4 PIPO (Parallel Input Parallel Output)



- Data loaded simultaneously.
- Output available on all lines simultaneously.
- **Typical application:** High-speed temporary storage.

II. CLASSIFICATION BASED ON APPLICATION

1 Shift Register



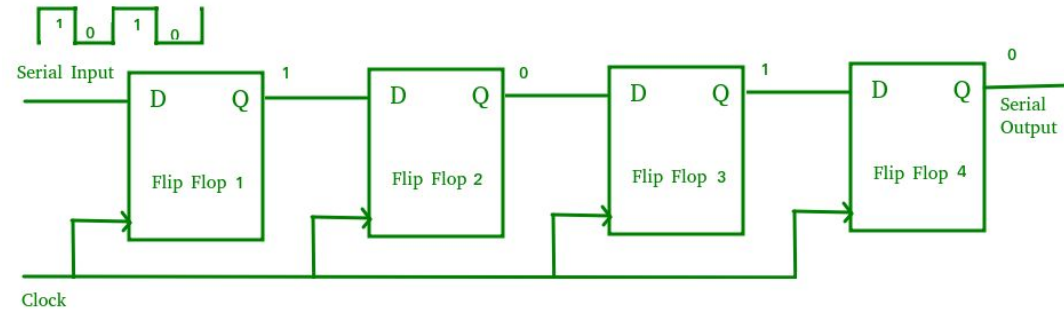
- Optimized for data shifting and bit manipulation.
- Performs logical and mathematical operations (e.g., multiplication/division by 2).
- Can be SISO, SIPO, or PISO.

2 Storage Register



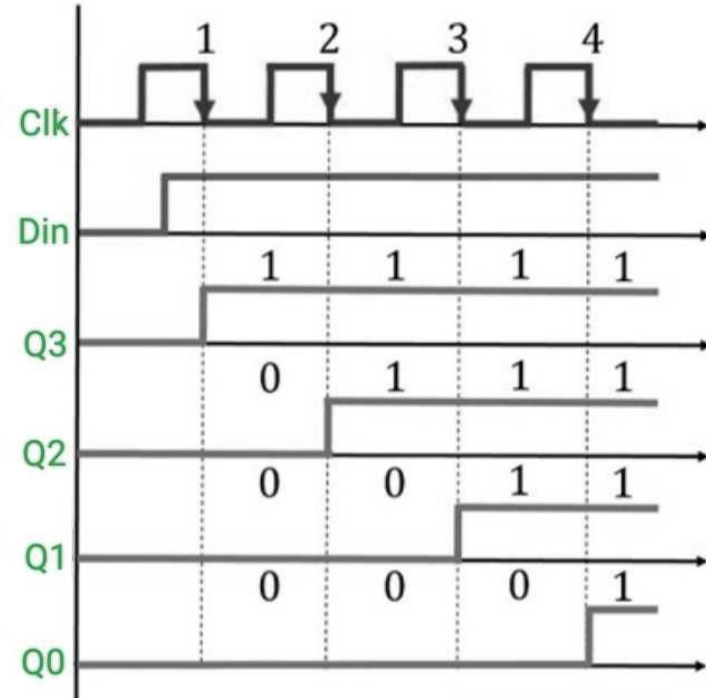
- Designed for temporary data retention without shifting.
- Example: A PIPO Register (e.g., Parallel Load Register).
- Maintains data until new data is loaded or power is removed.

Serial Input Serial Output (SISO)



Applications of SISO Shift Registers

- Data Storage and Retrieval
- Serial-to-Parallel Conversion
- Delay and Time-Sequence Generation
- Data Encryption and Decryption
- Frequency Division and Counting



Serial Input Serial Output (SISO)

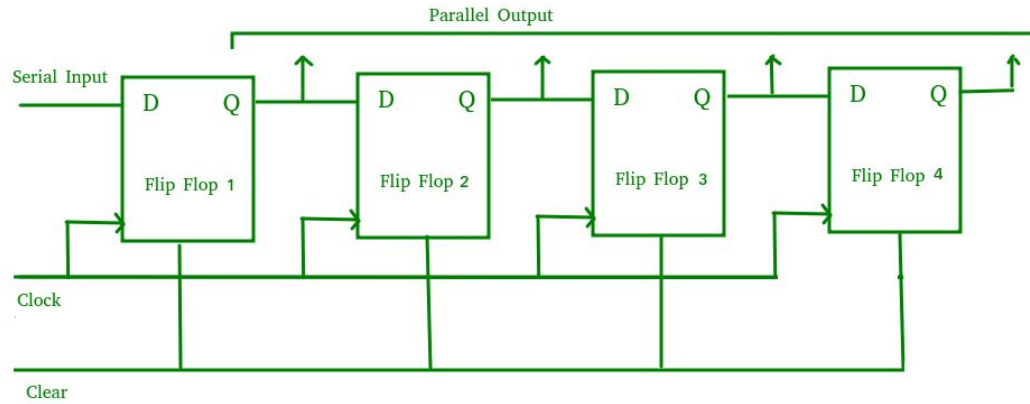
Advantages of SISO Shift Register

- Simplicity
- Space Efficiency
- Economical
- Sequential Processing
- Compatibility

Disadvantages of SISO Register

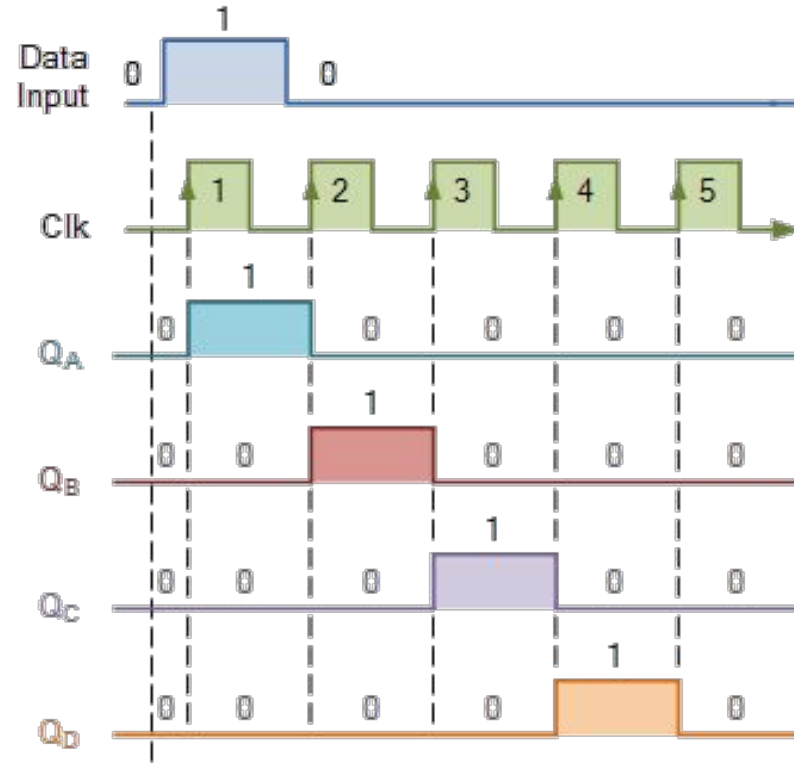
- Slower Data Transmission
- Limited Capability for Parallel Processing
- Complexity for Performing Multiple Operations
- Data Synchronization Difficulties
- High propagation delays

Serial Input Parallel Output (SIPO)

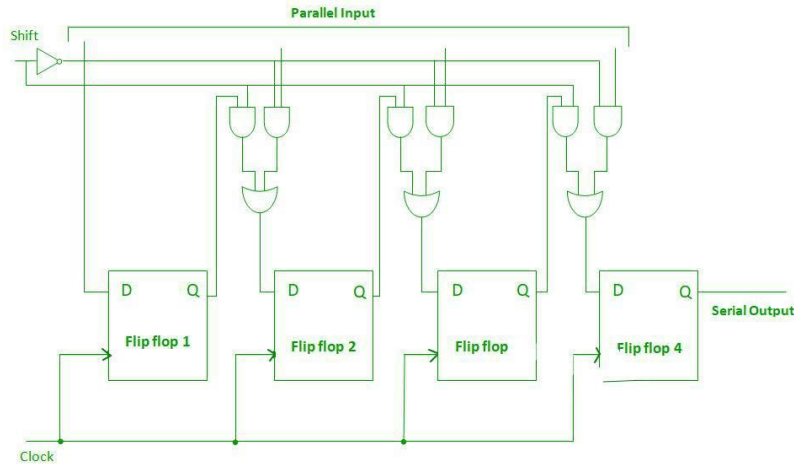


Applications of SIPO Shift Registers

- Data Storage and Buffering
- Data Storage and Buffering
- Address Decoding
- Control Systems



Parallel Input Serial Output (PISO)

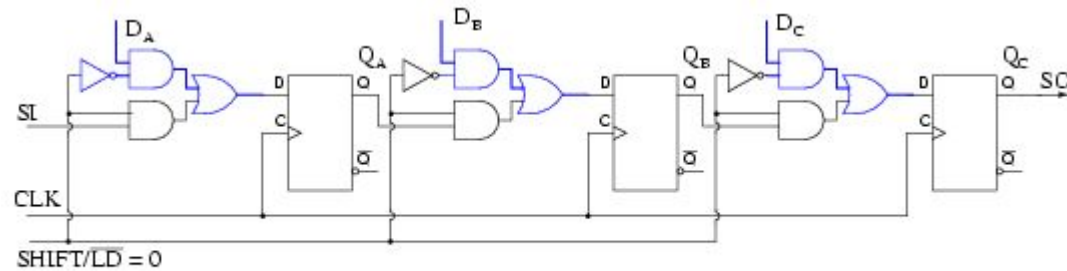


Applications of SIPO Shift Registers

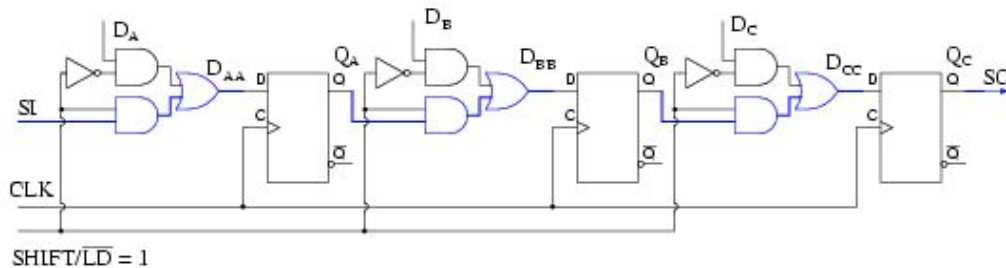
- Serial Data Transmission
- Sensor Data Acquisition
- Serial-to-Parallel Conversion
- Data Logging

Clock Cycle	SH / LD	Q_1	Q_2	Q_3	...	Q_{n-1}	$Q_n = \text{Data out}$	
1	0	B_1	B_2	B_3		B_{n-1}	B_n	Parallel data Loading Operation
2	1	0	B_1	B_2	...	B_{n-2}	B_{n-1}	
3	1	0	0	B_1	...	B_{n-3}	B_{n-2}	
...		...	...	...	...	...	...	Serial data retrieval
$n-1$	1	0	0	0	...	B_1	B_2	
n	1	0	0	0	...	0	B_1	

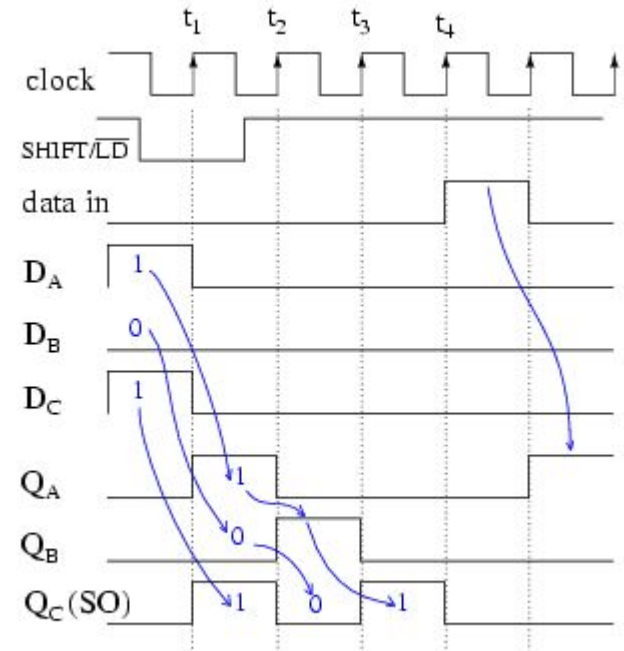
PISO with Serial Input



Parallel-in/ serial-out shift register showing parallel load path



Parallel-in/ serial-out shift register showing shift path



Parallel-in/ serial-out shift register load/shift waveforms

Parallel Input Serial Output (PISO)

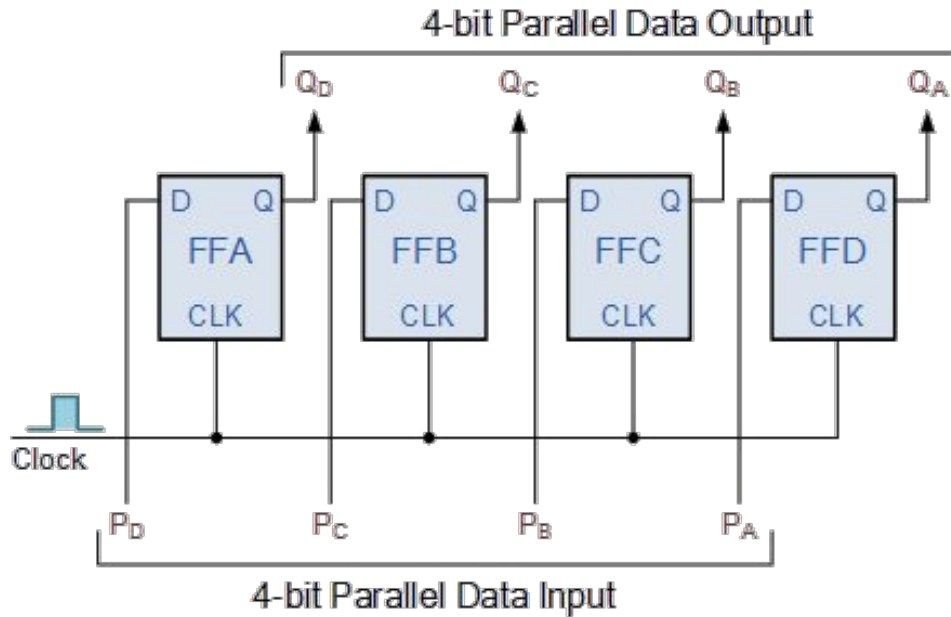
Advantages of PISO Shift Register

- Efficient Data Transmission
- Simplified Circuit Design
- Flexibility
- High-Speed Operation
- Cost-Effective

Disadvantages of PISO Shift Register

- Complex Control Logic
- Data Latency
- Limited Data Rate
- Error Accumulation
- Increased Circuit Complexity

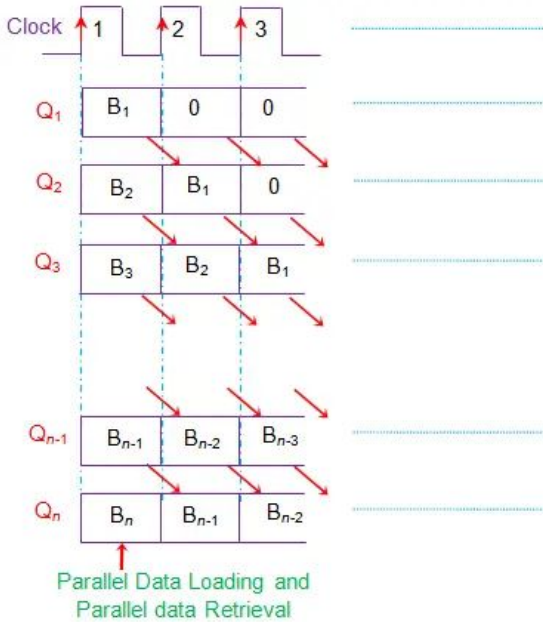
Parallel Input Parallel Output (PIPO)



Applications of PIPO Shift Registers

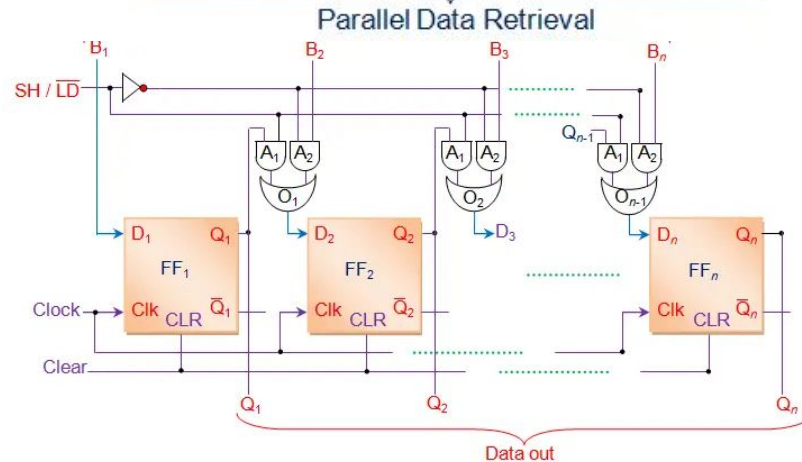
- Data storage
- Data Transfer
- Data Manipulation
- Serial to Parallel Conversion

PIPO Register with Shifting



Clock Cycle	SH / LD	Q_1	Q_2	Q_3	...	Q_{n-1}	Q_n
1	0	B_1	B_2	B_3	...	B_{n-1}	B_n
2	1	0	B_1	B_2	...	B_{n-2}	B_{n-1}
3	1	0	0	B_1	...	B_{n-3}	B_{n-2}
...	...	...	...	...	...	...	...

Parallel Data Loading ...



Parallel Input Parallel Output (PIPO)

Advantages of PISO Shift Register

- Specialization
- Flexibility
- Reduced Capital Investment
- Risk Management
- Leverage Supplier Expertise
- Shorter Lead Times
- High-Speed Data Transfer

Disadvantages of PISO Shift Register

- Higher Pin Count
- Potential for Signal Skew
- Increased Complexity for Large-Scale Designs
- Limited Flexibility

Counter

Counter is a sequential circuit which is used for a counting pulses, often in response to a clock signal.

Types of Counters based on Clock Signal Synchronization:

- Asynchronous Counter
- Synchronous or Ripple Counter

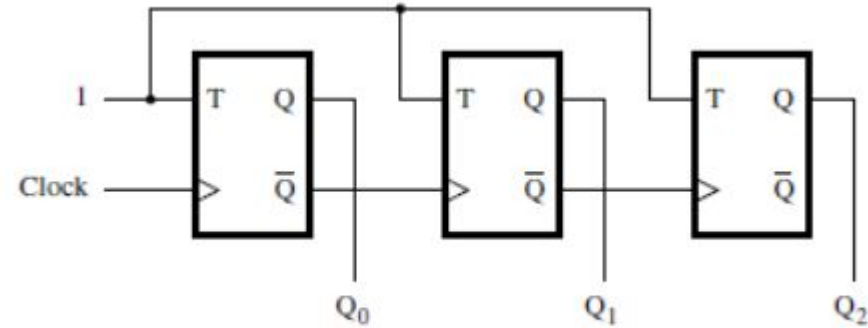
Types of Counters based on Counting Direction/Behavior:

- Up Counter
- Down Counter
- Up/Down Counter & more

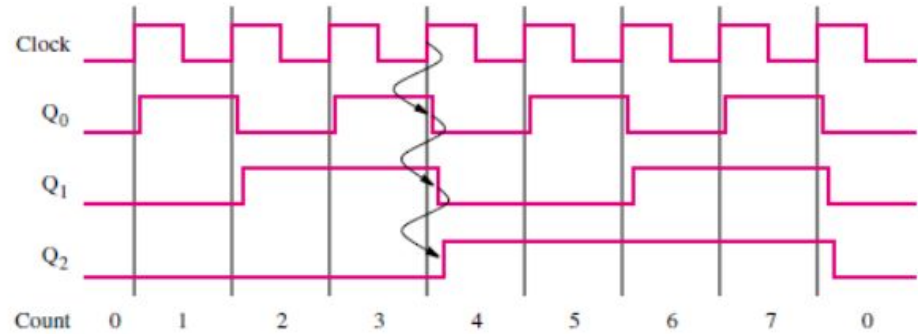
Asynchronous vs Synchronous Counters

Asynchronous Counter	Synchronous Counter
1. Flip flops are connected in such a way that the o/p of first flip flop drives the clock of next flip flop.	1. There is no connection between o/p of first flip flop and clock of next flip flop
2. Flip flops are not clocked simultaneously.	2. Flip flops are clocked simultaneously.
3. Circuit is simple for more number of states.	3. Circuit becomes complicated as number of states increases.
4. Speed is slow as clock is propagated through number of stages.	4. Speed is high as clock is given at a same time.

Asynchronous Up Counter: T-FF

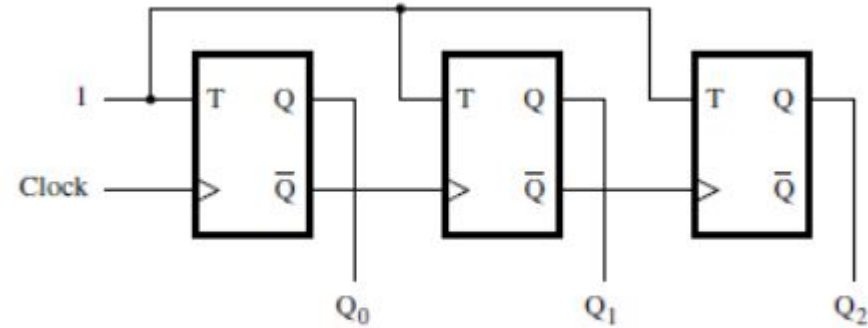


(a) Circuit

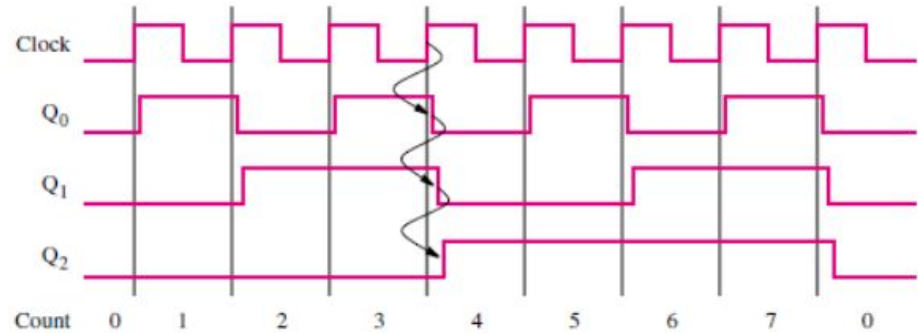


(b) Timing diagram

Asynchronous Up Counter: T-FF

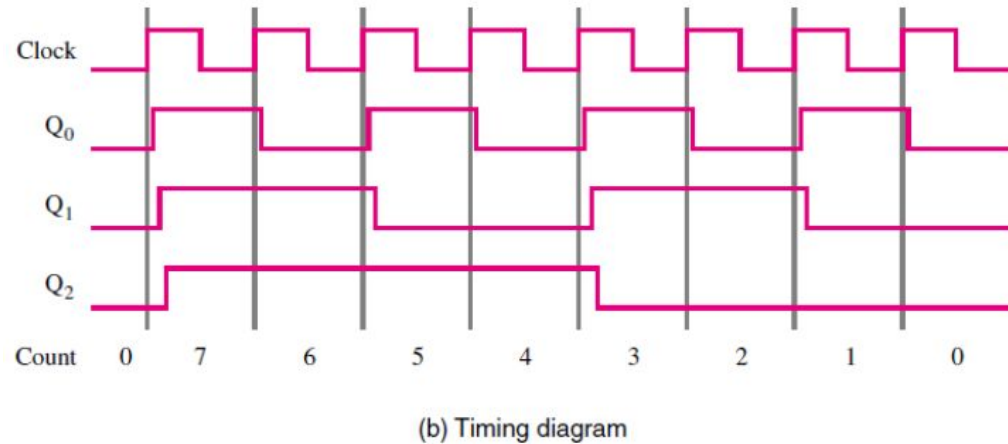
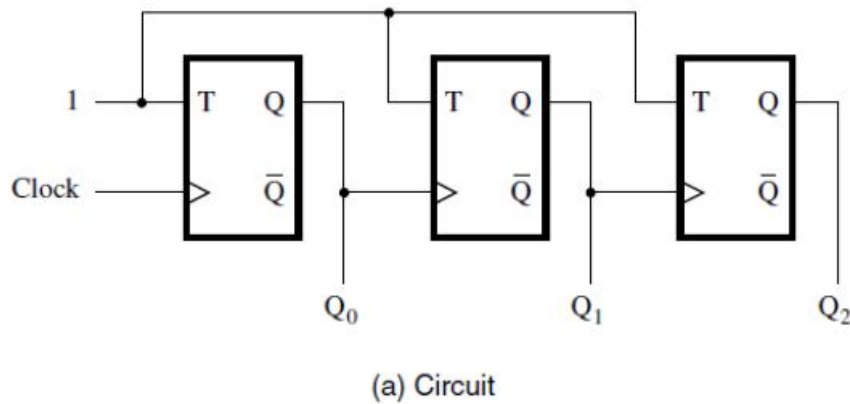


(a) Circuit

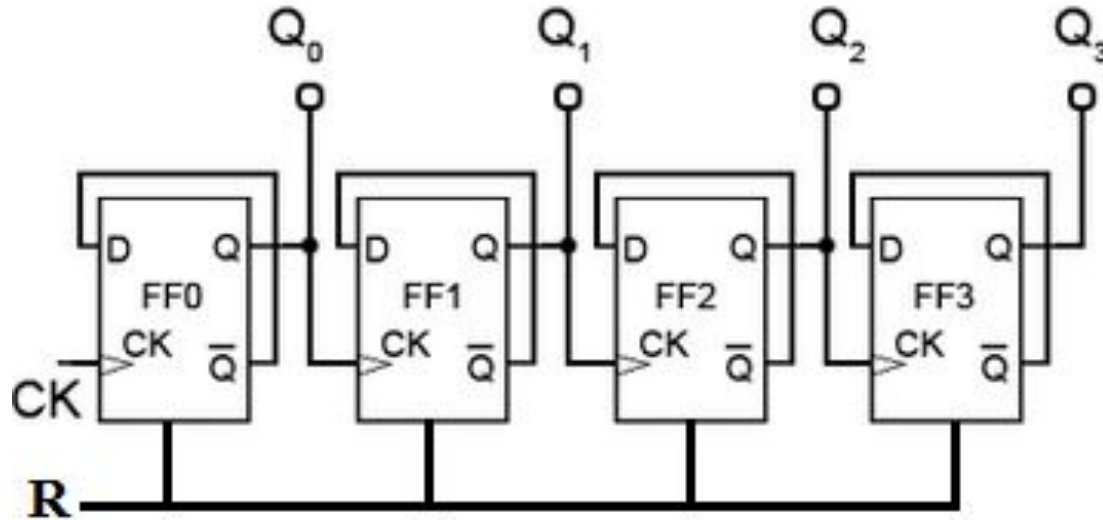


(b) Timing diagram

Asynchronous Down Counter: T-FF

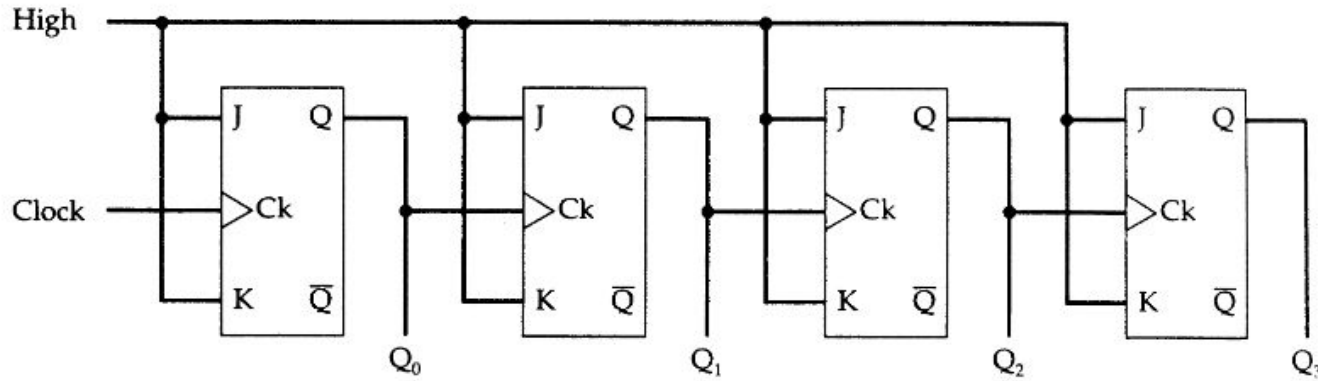


Asynchronous down Counter: D-FF



How can you design an asynchronous up counter using D flipflops?

Asynchronous down Counter: JK-FF



(a) Sequential Circuit

How can you design an asynchronous up counter using JK flipflops?

Excitation Table for Flip Flops

SR flip-flop				D flip-flop		
$Q(t)$	$Q(t + 1)$	S	R	$Q(t)$	$Q(t + 1)$	D
0	0	0	×	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	0
1	1	×	0	1	1	1

JK flip-flop				T flip-flop		
$Q(t)$	$Q(t + 1)$	J	K	$Q(t)$	$Q(t + 1)$	T
0	0	0	×	0	0	0
0	1	1	×	0	1	1
1	0	×	1	1	0	1
1	1	×	0	1	1	0

Designing Synchronous Counters

1. Identify the number of states. Such as: to count 0 to 7, you need 8 states.
1. For 2^n states, you need n-flip flops.
1. Construct the state-diagram: present states and future states.
1. Construct the excitation table of the available FF.
1. Make a connection between the state table and the excitation table.
2. Find a boolean expression for the inputs to the available FF in terms of present states.

Synchronous up-counter: T-FF

Table 5.1 Derivation of the synchronous up-counter.

Clock cycle	Q ₂	Q ₁	Q ₀
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Q₁ changes
Q₂ changes

$$T_0 = 1$$

$$T_1 = Q_0$$

$$T_2 = Q_0 Q_1$$

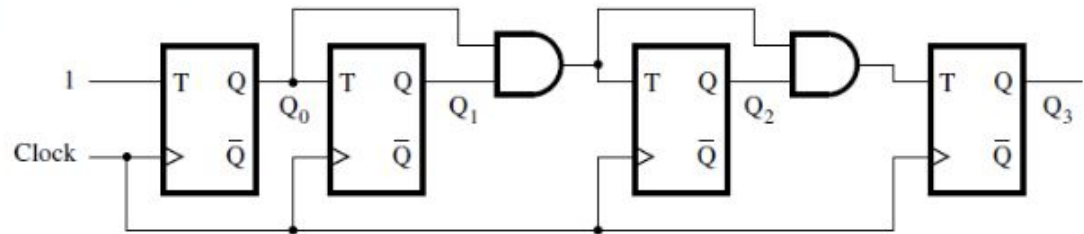
$$T_3 = Q_0 Q_1 Q_2$$

.

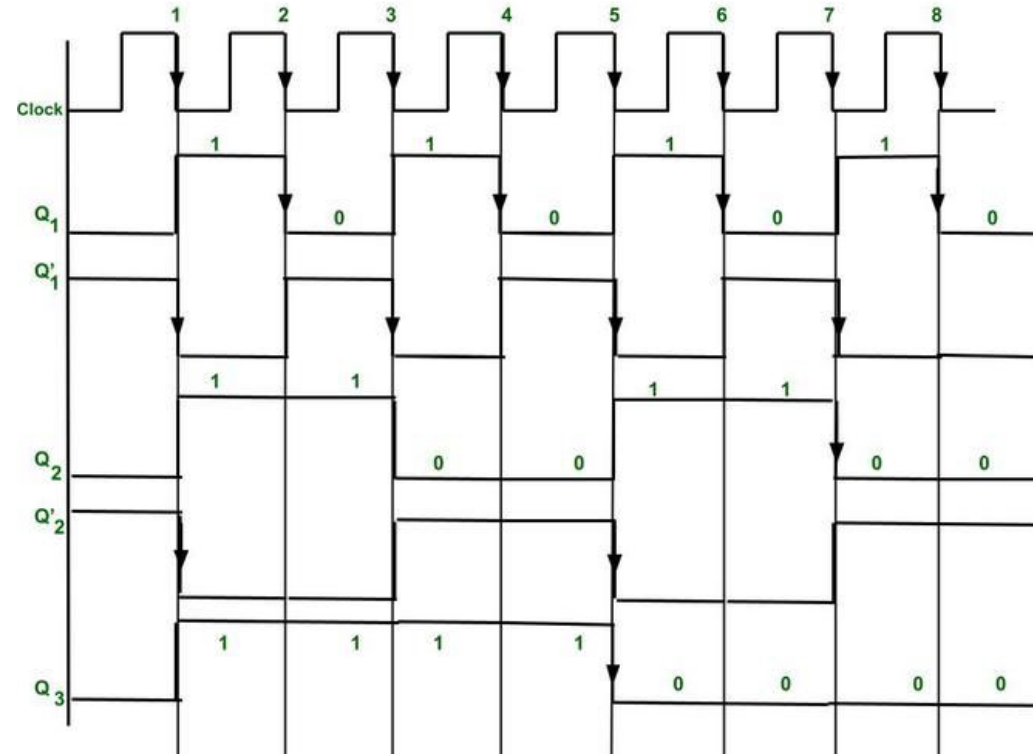
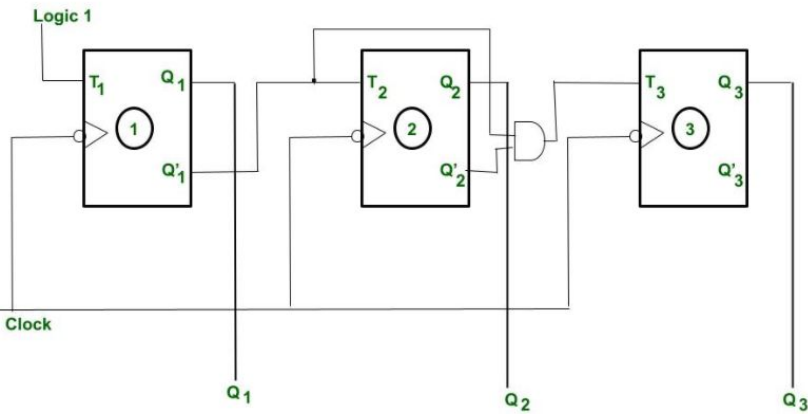
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.

$$T_n = Q_0 Q_1 \cdots Q_{n-1}$$



Synchronous down-counter: T-FF



Synchronous up-counter: D-FF

$$D_0 = Q_0 \oplus 1 = \overline{Q_0}$$

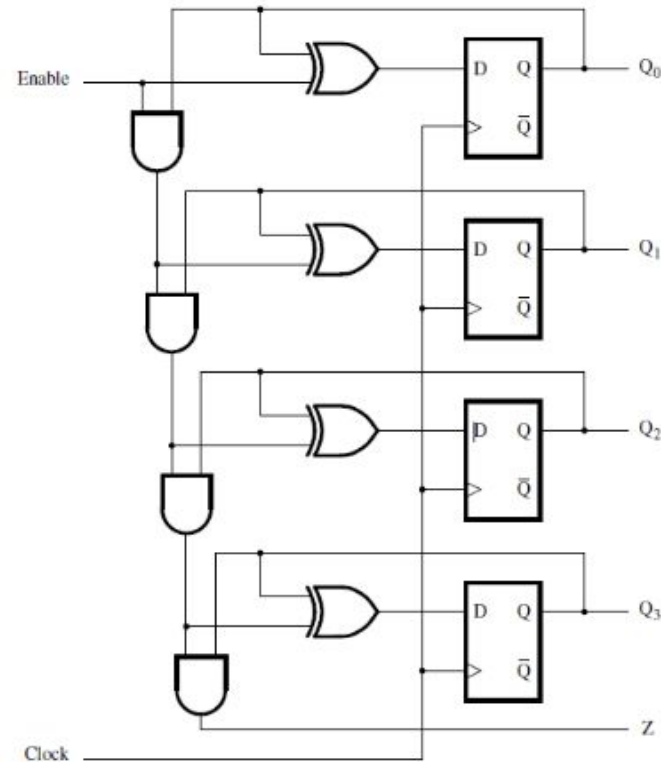
$$D_1 = Q_1 \oplus Q_0$$

$$D_2 = Q_2 \oplus Q_1 Q_0$$

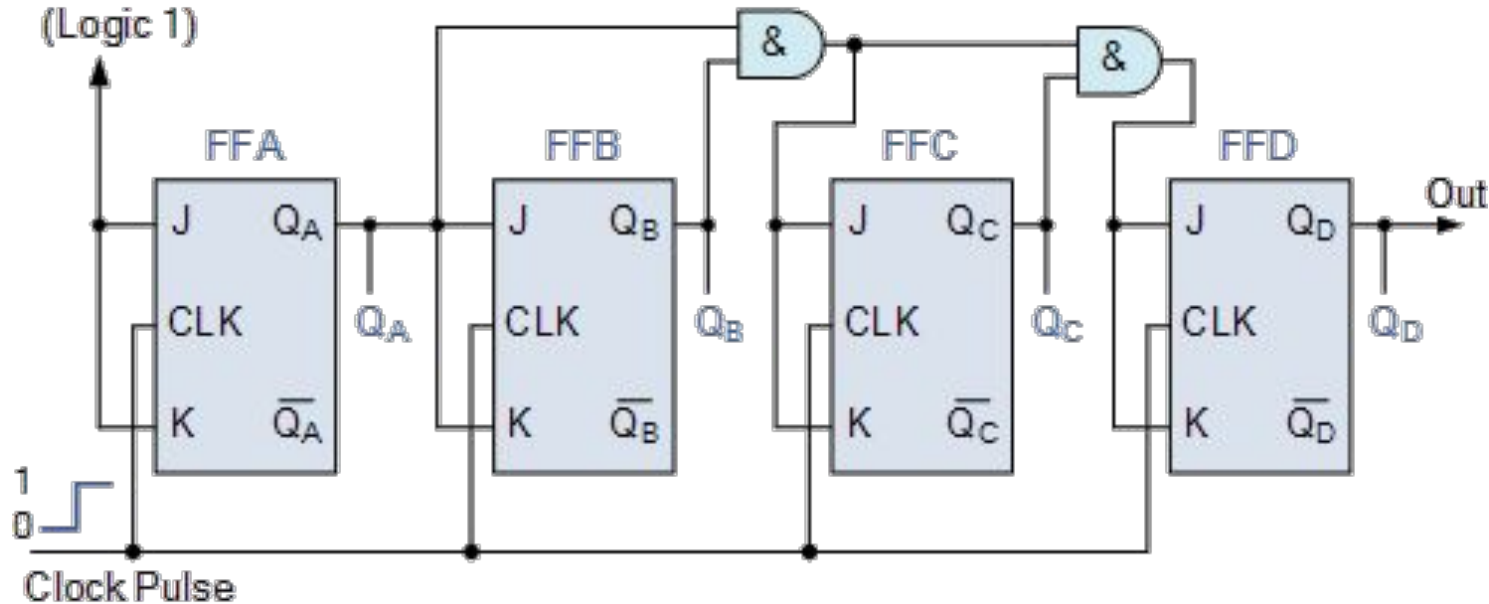
$$D_3 = Q_3 \oplus Q_2 Q_1 Q_0$$

$$D_i = Q_i \oplus Q_{i-1} Q_{i-2} \cdots Q_1 Q_0$$

**How can you design an
synchronous down counter
using D flipflops?**



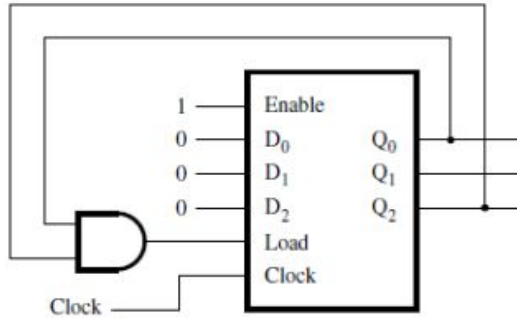
Synchronous up-counter: JK-FF



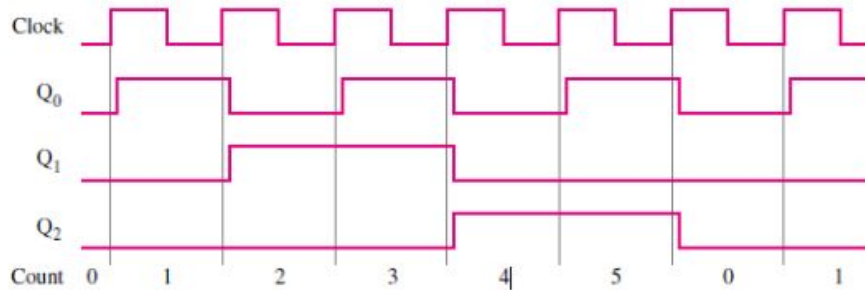
How can you design an synchronous down counter using JK flipflops?

Modulo N-counter

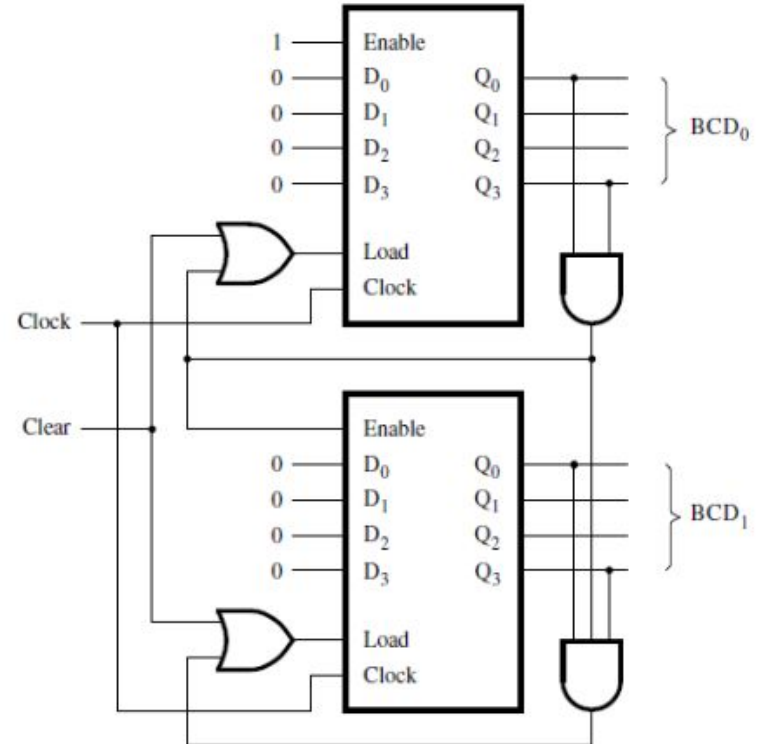
N = 6



(a) Circuit



2-digit BCD Counter



Ring Counter

Ring counter is a type of synchronous counter, typically built with flip-flops. It works by shifting a “1” through a series of flip-flops, one with each clock cycle. It is called a ring counter because the 1 loops back to the start once it reaches the end, forming a cycle.

A ring counter, has n unique states if n is the number of flip-flops used.

Ring Counter

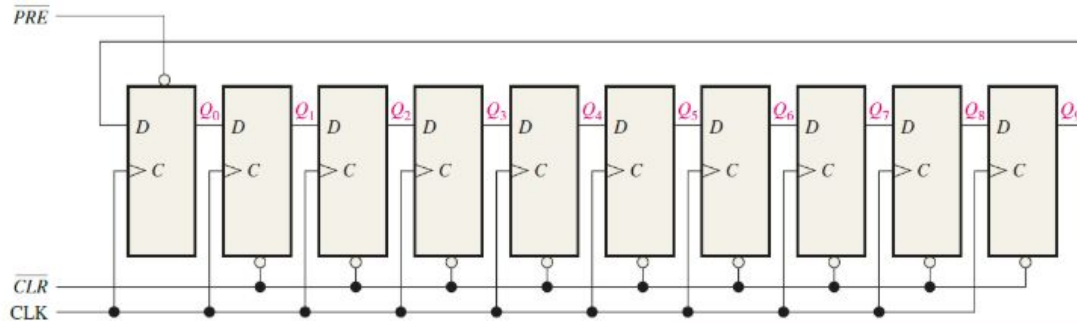


TABLE 8-5

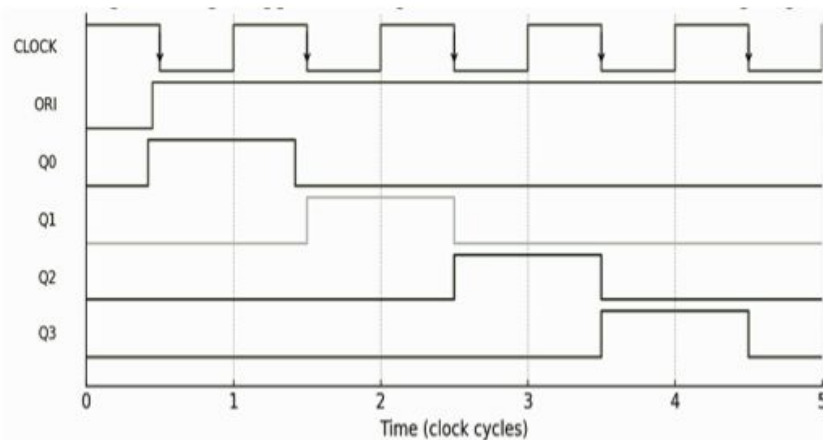
Ten-bit ring counter sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9
0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	0	0	0
4	0	0	0	0	1	0	0	0	0	0
5	0	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	0	1	0	0	0
7	0	0	0	0	0	0	0	1	0	0
8	0	0	0	0	0	0	0	0	1	0
9	0	0	0	0	0	0	0	0	0	1

Ring Counter

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	↓	0	1	0	0
high	↓	0	0	1	0
high	↓	0	0	0	1
high	↓	1	0	0	0

PRESETED 1



Initial state: 1000 (the first flip-flop is set to 1, and others are set to 0).

Clock cycle 1: The '1' bit shifts to the next flip-flop, resulting in the state 0100.

Clock cycle 2: The '1' bit shifts to the next flip-flop, resulting in the state 0010.

Clock cycle 3: The '1' bit shifts to the next flip-flop, resulting in the state 0001.

Clock cycle 4: The '1' bit shifts back to the first flip-flop, resulting in the state 1000.

Johnson Counter

Johnson counter (twisted/switch-tail ring counter) is a special type of ring counter. This counter connects the complement of the output of the last shift register to the input of the first register and circulates a stream of ones followed by zeros around the ring.

In the Twisted Ring Counter, **number of states = 2X the number of flip-flops.**

Johnson Counter

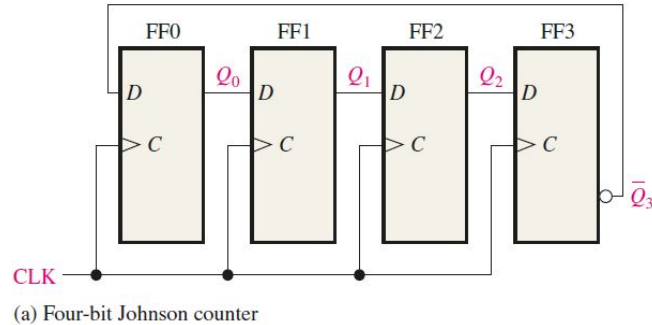


TABLE 8-3

Four-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

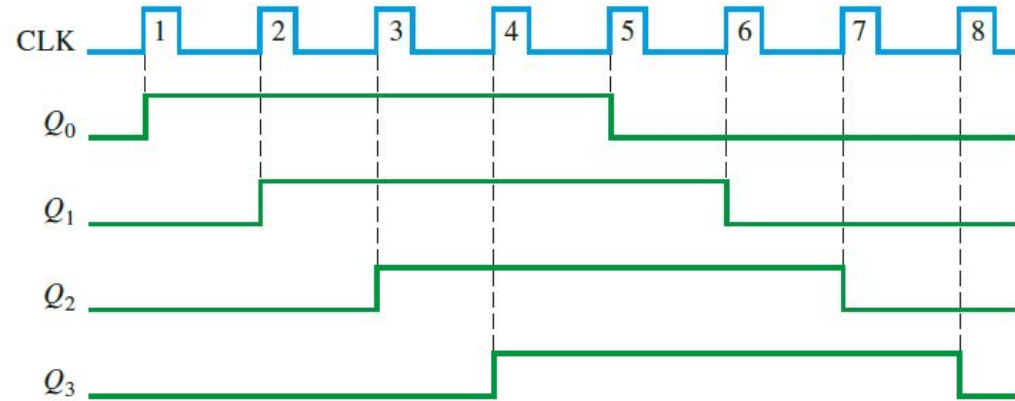


FIGURE 8-22 Timing sequence for a 4-bit Johnson counter.